How-To Library

Informatica

Migrating IDD Applications to the Business Entity Data Model

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Abstract

Before MDM Multidomain Edition version 10.0, you configured Informatica Data Director (IDD) to use the subject area data model. In version 10.0, IDD adopted the use of the business entity data model. The subject area data model differs significantly from the business entity data model, and the task to migrate an IDD application to the business entity data model might appear daunting. This article explains the main differences between the subject area data model and the business entity data model and explains how you can migrate IDD applications to the business entity data model.

Supported Versions

- MDM Multidomain Edition 10.2

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Overview

Informatica Data Director (IDD) was based on the subject area data model (SA data model). In version 10.2, IDD is based on the business entity data model (BE data model). After you upgrade to version 10.2, you can choose to migrate the IDD applications to the BE data model. This document summarizes the benefits of the BE data model, the differences between the data models, and describes how to migrate your IDD application to the BE data model.

Informatica recommends that you migrate to the BE data model, which offers multiple benefits over the SA data model. However, the move from subject areas to business entities is optional. Before you migrate the IDD application, especially if your business uses extensive customization and user exits, consider the impact of migration and review the benefits and differences of the different data models.

Benefits of the Business Data Model

Informatica recommends that you move to the BE data model because it offers the following benefits over the SA data model:

Integrated business entity services

A business entity service is a set of operations that run MDM Hub code to create, update, delete, and search for records in a business entity. You can develop a custom user interface that can run Java code or JavaScript code to make business entity service calls.

By using business entity services, which are generated directly from the BE data model, you can hide the complexity of the MDM Hub table structure from end users.

Configurable user interface

You can design user interface layouts that can contain standard and custom components, such as a Twitter feed. Business analysts can easily create layouts targeted at specific user roles.

Ability to merge child records

With subject areas, you cannot merge child records. With business entities, you can merge child records that are at the same descendant level.

Business entity views

Business entity views represent a condensed version of a business entity. You can simplify integration with existing systems by using business entity views to create a view that matches your existing processes.

Integration with Data as a Service (DaaS) providers

You can augment your business entity data with data from third-party data providers. You can integrate with DaaS providers to access data that is reliable, accurate, and complete.

Unlimited nesting of descendant records

A subject area can include descendant records up to the grandchild level. Business entity models can have an unlimited depth of descendant records.

Improved user interface

The Entity 360 framework screens (E360 views) are cleaner, more colorful, and more intuitive.

Simplified and enhanced data search

A smart search on business entities retrieves data faster than a faceted search on subject areas.

Overview

There are multiple differences between the BE data model and SA data model, including differences in terminology and functional views. Although the BE data model offers many benefits, there is also some loss of functionality in a move to the BE data model.

The following table describes some of the differences between the SA data model and the BE data model:

Subject Area Data Model	Business Entity Data Model
Configured through the IDD Configuration Manager	Configured through the Provisioning tool
Supports user exits	Does not support user exits. Instead, use DaaS provider configurations to replace user exits. Create external services and call them from internal business entity services.
Supports Import Master Data	Not available
Supports masking data	Does not support masking data. Instead, views can be created that users cannot access. Also, Dynamic Data Masking (DDM) is integrated with the MDM Hub. You configure DDM through the Hub Console.
Tree unmerge and linear unmerge	Tree unmerge only
Supports Jaspersoft reports	Jaspersoft reports only work in a Start page if it is the only component in the Start page. Informatica recommends to use high charts instead of Jaspersoft reports.
Ability to override merges	Cannot override merges. Instead, merge records and then use the Cross-reference Records view to choose the correct merge.

Data Model

The BE data model differs significantly from the SA data model.

Subject Area Data Model

In the SA data model, data is organized around subject areas and aggregated into subject area groups. A subject area represents a collection of data that can be treated as a unit from a business perspective.

You configure lookups in a physical lookup base object table and IDD used metadata or configuration files to populate the lookup values.

You configure relationships within subject areas that are based on the relationships that exist between base objects in the Hub Store. For information about subject area configuration, see the *Informatica MDM Multidomain Edition IDD Implementation Guide*.

Business Entity Data Model

In the BE data model, data is organized around business entities. Business entities represent entities with significance to an organization. Organizations commonly define business entity types to represent customers, suppliers, employees, products, and accounts.

The business entity model is a tree model of nodes. Each node corresponds to a base object table in the MDM Hub. Each field corresponds to a column of a base object table in the MDM Hub. The business entity

model is based upon a root node. The root node is synonymous with the business entity. For example, a business entity model with a Person root node is considered to be a Person business entity.

After the root node is established, you can use the Provisioning tool to create child nodes that have a one-to-one or one-to-many relationship with their parent nodes. You can also use the Provisioning tool to configure reference entities, which are business entities that are associated with lookup base objects. For information about the Provisioning tool, see the *Informatica MDM Multidomain Edition Provisioning Tool Guide*.

Configuration Tool

For the SA data model, to configure an IDD application, you use the IDD Configuration Manager. For the BE data model, to configure an IDD application, you use the Provisioning tool.

IDD Configuration Manager

The IDD Configuration Manager is a web-based utility used to add, change, and manage IDD applications that are based on the SA data model. An IDD application consists of an XML configuration file, resource bundles, help files, and other components. A complete IDD application can be imported or exported as a ZIP file containing all of these components.

The IDD Configuration Manager does not expose all available configuration options. To configure some functionality, such as custom components, you must export and edit the XML configuration file directly, before you import the file back into the IDD Configuration Manager.

For more information about the IDD Configuration Manager, see the *Informatica MDM Multidomain Edition IDD Implementation Guide*.

Provisioning tool

The Provisioning tool is a web-based utility that you use to create and manage business entity data models and the related IDD E360 applications. You can use the Provisioning tool to define business entity models, tasks, and transformations, and design the user interface for Informatica Data Director.

You can define and design the following items:

- Business entities
- Business entity views
- Reference entities
- Relationships
- Transformations
- Task configurations
- Extensions, such as SOAP services
- User interface components
- User interface layouts

For more information about the Provisioning tool, see the *Informatica MDM Multidomain Edition Provisioning Tool Guide*.

Informatica Data Director

Informatica Data Director has new modes, views, and search functionality.

Modes for Informatica Data Director

Informatica Data Director now has three modes. After you upgrade the MDM Multidomain Edition software, IDD continues to display the SA-based legacy views and features. After you migrate an IDD application to the business entity data model, you switch to the BE-based E360 views and features.

The following table describes the different modes:

Mode	Description	User Interface
Legacy Mode	If you do not migrate your application to the BE data model, use the Legacy mode. You continue to see the SA-based features and views in IDD. You continue to use the IDD Configuration Manager to manage your applications. **Note:** If your legacy IDD application includes workflows, you must generate the BE schema as a requirement for Informatica Data Director to manage the workflow tasks. However, you continue to use the SA data model. You have not migrated to the BE data model.	The Legacy mode displays the following interface elements: - **Create in Data View** menu - **Queries** menu - **Start** page - **Tasks** tab - **Data** tab for legacy views - **Search** tab for search query results - Links to the **Data View** from menus on other views - Custom tabs, if configured
E360 Mode	If you migrate to the BE data model, select the E360 mode. Business users can use the BE-based features and E360 views. You use the Provisioning tool to manage your migrated applications. **Note:** The E360 mode is the default mode for new customers. New customers start with the BE data model and the E360 views.	Displays the following interface elements: - **Create** menu - **Search** box - **Start** page, which you design - **Search** tab for smart search results - **Task Manager** tab - Entity tabs for E360 views. The tab name is configurable, but is usually the name of the opened business entity. - Links to the **Business Entity** view from menus on other views - Custom tabs, if configured
Hybrid Mode	If you migrate to the BE data model, but you want to access to all the SA-based features and views plus the BE-based features and views, use the Hybrid mode. **Caution:** If you need to make a change to the schema while in Hybrid mode, you must make the change twice: first in the IDD Configuration Manager and again in the Provisioning tool. Informatica recommends that you use the Hybrid mode only on a temporary basis.	Displays all the interface elements from the Legacy mode and the E360 mode.

This document assumes that you are migrating to the E360 mode. If you want to use the Legacy mode or the Hybrid mode, see "IDD Modes FAQs" on page 22 before you begin the migration.

Views

The Entity 360 framework (E360) views are similar to the legacy views. In some cases, the E360 views have different behavior or offer some enhancements.

Note: In comparing the views, we assume that you are already familiar with the behavior of the legacy views.

The following table describes the similarities between the views and then highlights how the E360 views are different:

Legacy Views for Subject Area Model	E360 Views for Business Entity Model	Similarities Between the Views	Differences for the E360 Views
Data View	Business Entity	Both views display master data for the selected business entity.	In the Business Entity view, you can add related social information, such as the entity's Twitter feed. You can also add helpful components, such as a component that displays similar business entities. You can configure more than one Business Entity view, and add the custom views to the Views list.
XREF View	Cross-reference Records	Both views display the cross-reference records that contain data from the source systems. You can create the best version of the truth.	The Cross-reference Records view includes all levels of descendant records.
Matches View	Matching Records	Both views display records that satisfied a match rule.	In the Matching Records view, users can match and merge descendant records. Users can also run a smart search to find and add similar records. You cannot manually override a value in the Matching Records view.
History View	History	Both views display data events in chronological order.	In the E360 version of the History view, the events appear in a vertical chronology, and data change event details appear in a panel to the right.
Effective Periods View	Timeline	Both views display information about data events with effective periods along a timeline.	n/a
Hierarchy View	Hierarchy	Both views display relationships for a selected business entity.	The E360 version of the Hierarchy view does not permit the following actions: - Finding a duplicate entity. Use a Business Entity view with the Similar Records component, or use the Matching Records. - Initiating a merge. Use the Matching Records view for this task. - Sharing a bookmark URL. You can still share an image of a hierarchy.

Search

With the subject area data model, you use search queries to find master data. With the business entity data model, you use smart search to find business entities.

Search Queries for Subject Areas

You create a search query to search selected fields for specified field values. First you select fields from the subject area that you want to search. For each field, you enter a value that you want to search for within the field.

Smart Search for Business Entities

Smart search is a full text search that runs on business entities. In the business entity models, you identify the fields that you want to be searchable. In an IDD application, users can type their search term in a Search box. Unless they restrict the search, the results include all business entities that contain that term in any of their searchable fields. Smart search is faster than search queries.

Overview

This procedure assumes that, after you migrate to the business entity data model, you want to configure the IDD application to use E360 mode.

Note: If you want to use the Legacy mode or the Hybrid mode, see "IDD Modes FAQs" on page 22 before you begin the migration.

Before you begin, ensure that you have the prerequisite software configured.

Prerequisites

Before you migrate to the BE data model and the E360 views, configure the software to support smart search and business process management (BPM) workflows. The software is installed as part of the installation or upgrade process. You need to ensure the software is configured correctly.

The following table lists the prerequisite software and tells you where to find the configuration instructions:

Software	Description	Instructions
Apache Solr and Apache ZooKeeper	The MDM Hub uses Apache Solr to perform smart searches, that is, full text searches on selected fields in the master data. Apache Solr is an open-source enterprise search application. SolrCloud uses a cluster of Solr servers to provide distributed indexing and search. Apache ZooKeeper is a centralized service that synchronizes searches across multiple Solr servers.	For information about configuring smart search, see the *Informatica MDM Multidomain Edition Configuration Guide*
ActiveVOS	Required when you want to use BPM workflows with the BE data model. The MDM Hub uses ActiveVOS to run BPM workflows and to create review tasks. When you upgrade, install ActiveVOS as part of the Hub Server upgrade process, and select the BE ActiveVOS workflow adapter as a workflow engine in the MDM Hub Console.	For information about installing and configuring ActiveVOS, see the *Informatica MDM Multidomain Edition Upgrade Guide* or the *Informatica MDM Multidomain Edition Installation Guide*, as applicable.

Prepare the ORS and the IDD Application

Ensure that your Operational Reference Store (ORS) and the IDD application meet the requirements for the business entity model.

The following table describes the requirements:

Area of Impact	Requirement	Description
Hierarchy entity base object	Foreign key relationships must map to a CODE column	If a foreign key maps to another type of column, remap it to a CODE column.
Subject area names	No underscores or other special characters in names	If some subject area names contain the underscore character or special characters, rename the subject areas to remove the underscore.
Validation	The legacy IDD application must be valid.	In the IDD Configuration Manager, validate the IDD application and resolve any errors.

Change the Mode for IDD

After you upgrade the MDM Multidomain Edition software, if you open an IDD application, by default you see the legacy views and search query feature. After you migrate, enable the E360 mode to see the E360 views and the smart search box.

To change the mode for IDD, you set properties in the MDM Hub Server properties file and the Process Server properties file.

1. Navigate to the following directory:

   `<MDM Hub installation directory>/hub/server/resources`

2. Open the `cmxserver.properties` file.
3. Set the properties to reflect the views that you want to show in the migrated IDD application.

   The following table lists the properties and the before and after settings:

Properties	Before = Legacy Mode	After = E360 Mode
cmx.dataview.enabled	true	false
cmx.e360.view.enabled	false	true
cmx.e360.match_xref.view.enabled	false	true
cmx.ss.enabled	false	true

4. Navigate to the following directory:

   `<MDM Hub installation directory>/hub/cleanse/resources`

5. Open the `cmxcleanse.properties` file.
6. Set the `cmx.ss.enabled` property to match the value that you set in 3.
7. After you save the changes to the properties files, restart the application server.

Generate the Business Entity Schema

You use the IDD Configuration Manager to generate the business entity schema.

During the generation process, the MDM Hub performs the following migrations:

- Creates a business entity model for each subject area
- Converts lookups to reference entities
- Migrates relationships

The process saves the business entity schema in the C_REPOS_CO_CS_CONFIG repository table.

1. Log into the IDD Configuration Manager.
2. Select the application to migrate.
3. In the **Applications** screen, click **Generate Business Entity Schema**.

   A message confirms that the schema was created. You can view the generated schema in the Provisioning tool.

Log in to the Provisioning Tool

You use the Provisioning tool to view and enhance the schema.

1. Open a new browser tab and use the same IP address and port number that you use for the IDD Configuration Manager:

   ```
   <IP address>:<port>/provisioning/
   ```

2. Enter the same user credentials that you use for the IDD Configuration Manager.

Create an E360 Application for IDD

Each Operational Reference Store (ORS) database can have one E360 application. In the Provisioning tool, create an E360 application that is based on the same ORS as the legacy application. For convenience, you might want to give it the same name as the legacy application.

Note: If you plan to use Hybrid mode, you must ensure that the E360 application name is the same as the legacy application name.

1. In the Provisioning tool, from the **Database** list, select the database with which you want to associate the application.
2. Click **UI Configuration** > **Application Editor**.

   The **Application Editor** appears.
3. Click **Create**.
4. In the **Properties** panel, specify the following application properties:

Property	Description
Name	Name of the application that appears in the Applications panel.
Display Name	Name of the application that appears in Informatica Data Director.

Property	Description
Source System	Source system with which you want to associate the application.
Session timeout (minutes)	Time in minutes to wait before an idle Informatica Data Director session can time out.

5. Click **Apply**.

   The application that you created appears in the **Tree View** panel and in the **Applications** panel.

6. Publish the changes to the MDM Hub.
   a. Click **Publish**.

      A change confirmation dialog box appears that prompts you to confirm the changes.

   b. Review the changes, and click **Confirm**.

      A validation process runs on the application. A confirmation dialog box appears that prompts you to publish the changes.

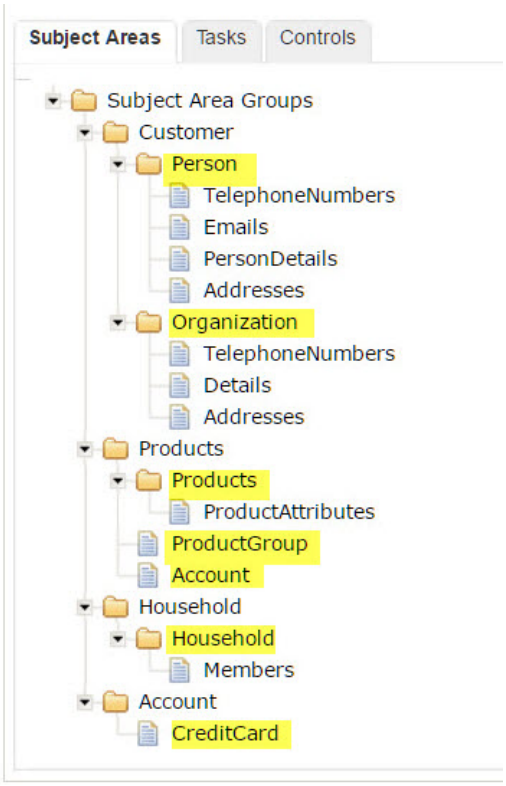
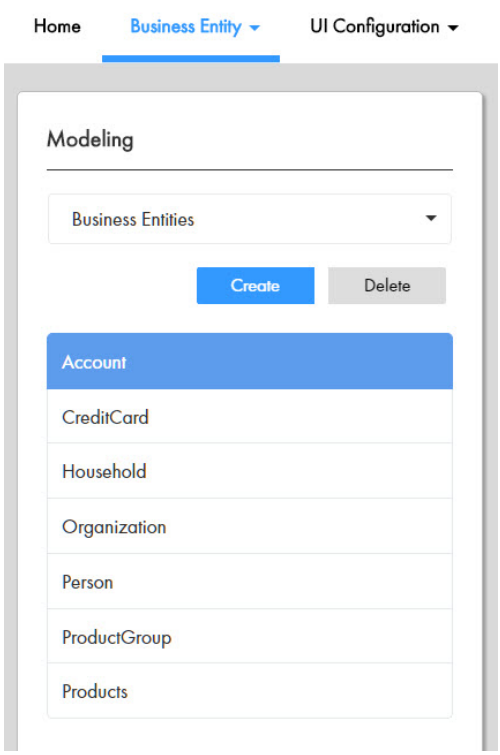
   c. Click one of the following buttons:
      - **Publish**. Saves the changes to the MDM Hub.
      - **No**. The changes remain in the temporary workspace.

Review the Generated Business Entity Schema

You use the Provisioning tool to view and enhance the schema in a user-friendly interface.

1. In the Provisioning tool, select the E360 application for the database.
2. Click **Business Entity** > **Modeling**.
3. Select **Business Entities** from the drop-down list.

The following images show the original list of subject areas and the generated list of business entity models:

IDD Configuration Manager Subject Areas	Provisioning Tool Business Entity Models
	

4. To see the lookups, select **Reference Entities** from the drop-down list.
5. To see the relationships, select **Relationships** from the drop-down list.

Enable Smart Search

To enable smart search, you open the business entity models and select the fields that you want to be searchable. When you are done, you run the **Initially index Smart Search data** batch job from the MDM Hub Console.

Caution: Do not mark the rowidObject field as searchable. The business entity data model does not support searches on the rowidObject field.

Run the batch job on all business entity models.

1. In the Provisioning tool, click **Business Entity** > **Modeling**, and select **Business Entities**.
2. Click a business entity model.

   The business entity model opens.
3. In the business entity panel, expand the **field** folder.

   All the fields are displayed
4. For each field that you want to be searchable, select the field and click **Searchable** in the Properties panel.
5. After you identify the searchable fields in the business entity models, click **Publish**.

Attention: You must publish the searchable fields to the database. If you do not publish the changes, the next step fails because there are no searchable fields to index.

6. In the MDM Hub Console, run the **Initially index Smart Search data** batch job on the business entity models that contain searchable fields:
   a. Log in the Hub Console and select the Operational Reference Store database.
   b. In the **Utilities** workbench, click **Batch Viewer**.
   c. Select **Group By Procedure Type**.
   d. Find **Initially index Smart Search data**.
   e. Select each business entity model and click **Execute Batch**.

      The index batch jobs run.

      Tip: Alternatively, you can create a batch group, add all the index jobs for the business entity models that you want to index, and then run the batch group.

Design a Start Page

You design the Start page for the E360 application in the Layout Designer. In the Start page, add components, such as a task inbox, charts, social media feeds, and other external resources.

In some cases, you need to design the components before you can add them to the Start page. For more information about components and layouts, see the *Informatica MDM Multidomain Edition Provisioning Tool Guide*.

1. In the Provisioning tool, click **UI Configuration** > **Layout Designer**.

   The **Layout Designer** appears.
2. Click **Create** > **Start Page**.

   The **Define Layout Properties** page appears.
3. To define and identify the layout that you want to design, specify the following properties:

Property	Description
Layout name	Label for the layout that you design. After you design and publish the layout, the label appears in the layout list in the Layout Designer.
Unique label	Label used for generating a system ID.
Description	Optional. A meaningful description to identify the layout.
List option name	Label for the option that you want in the Informatica Data Director views list.

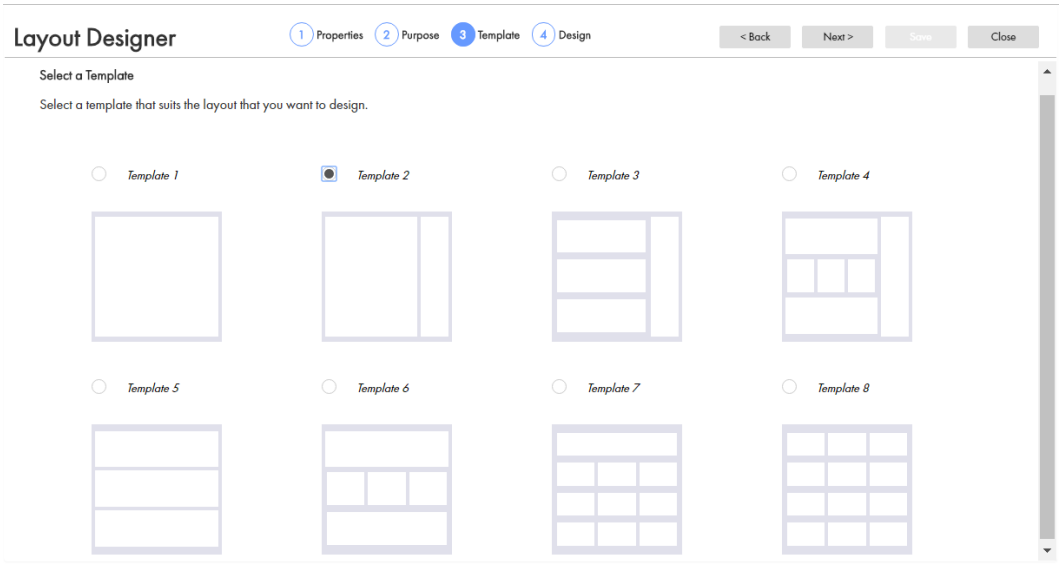
4. Click **Next**.

   The **Define Purpose** page appears.
5. Select the user roles that can access the layout.

   The user roles that appear on the **Define Purpose** page are configured in the MDM Hub environment.
6. Click **Next**.

   The **Select a Template** page appears.

The following image shows the **Select a Template** page:



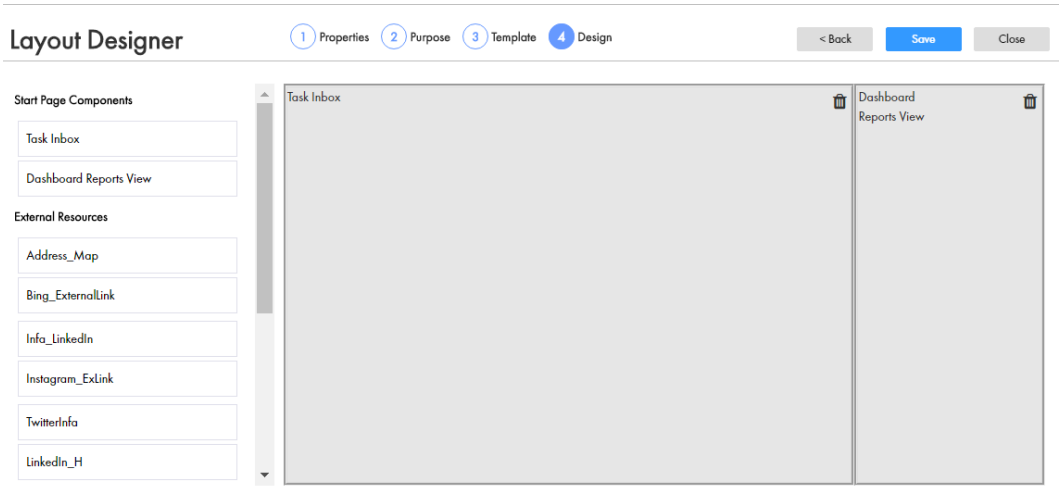
7. Select a template that is suitable to display the components you want in the Start page, and click **Next**.

   The **Design** page appears.

8. Drag the components that you want in the Start page layout into the workspace.

   For example, drag the **Task Inbox** and the **Dashboard Reports View** components into the workspace.

   The following image shows the **Design** page with the **Task Inbox** and the **Dashboard Reports View** components in the workspace:



9. Click **Save**.

   The Start page that you designed appears under the Start page layout category.

10. Publish the changes to the MDM Hub.

    a. Click **Publish**.

       A change confirmation dialog box appears that prompts you to confirm the changes.

    b. Review the changes, and click **Confirm**.

A validation process runs on the application. A confirmation dialog box appears that prompts you to publish the changes.

c. Click one of the following buttons:

- **Publish**. Saves the changes to the MDM Hub.
- **No**. The changes remain in the temporary workspace.

Next Steps

If you are implementing the BPM workflow and tasks, the last mandatory step is to set up the ActiveVOS workflow adapter, define task triggers, and process existing tasks. See the next section.

In future, you use the Provisioning tool to update and customize your application.

You can customize the E360 application in the following ways:

- Design Entity view components and layouts
- Configure cleanse transformations
- Create business entity views that contain a subset of fields
- Create new business entity models
- Add extensions for services, such as third party data providers

For more information about customizing an E360 application, see *Informatica MDM Multidomain Edition Provisioning Tool Guide*.

For more information about how the functionality in IDD Configuration Manager maps to the Provisioning tool, see "Migration FAQs" on page 21.

Overview

When you migrate Informatica Data Director to use the BE data model, you should migrate from the Siperian or SA ActiveVOS workflow adapter to the BE ActiveVOS workflow adapter.

The BE ActiveVOS workflow adapter uses the BE ActiveVOS workflows. The BE ActiveVOS workflows operate on business entities. Some screens, such as the E360 Matching Records view, require the BE ActiveVOS workflows to support child entity merges.

The MDM Hub supports a primary workflow adapter and a secondary workflow adapter. The secondary adapter allows you to process existing tasks with the old workflow adapter while you migrate to the BE ActiveVOS workflow adapter. When you migrate workflow adapters, set the BE ActiveVOS workflow adapter as the primary adapter.

Informatica Data Director launches the appropriate screens to process tasks. For example, if an unmerge task is created prior to migration, the legacy XREF view is launched when the task is processed.

Update the IDD Configuration for the Siperian Workflow Adapter

To view Siperian workflow adapter tasks in Task Manager, update the task configuration in the Informatica Data Director configuration file.

1. Update the Siperian BPM task type configuration.
   - Change `defaultApproval="true"` to `defaultApproval="false"`.
   - Set `creationType` to `NONE`.
2. Add the task configuration for the ActiveVOS tasks.

The following code sample shows how to configure business entity-based ActiveVOS tasks in the Informatica Data Director configuration file:

```
<tasks includeUnassignedTasks="true">
  <!-- Task Definitions -->
    <taskType taskTypeId="BeMergeTask" name="AVOSBeMerge" displayName="Merge"
    creationType="MERGE" displayType="MERGE">
    <description>Merge two records together.</description>
  </taskType>

  <taskType taskTypeId="BeUnmergeTask" name="AVOSBeUnmerge"
    displayName="Unmerge" creationType="UNMERGE" displayType="UNMERGE">
    <description>Unmerge an XREF record from a Base Object record.
    </description>
  </taskType>

  <taskType taskTypeId="BeOneStepApprovalTask" name="AVOSBeFinalReview"
displayType="NORMAL"
    displayName="Final review" creationType="NONE" pendingBVT="true">
    <description>Update a record and require the user to go through an
    approval process before completing the task.
    </description>
  </taskType>

  <taskType name="AVOSBeNotification" displayName="Notification"
    creationType="NONE" displayType="NORMAL">
    <description>Notification step in the workflow</description>
  </taskType>

  <taskType taskTypeId="BeTwoStepApprovalTask" name="AVOSBeReviewNoApprove"
displayType="NORMAL"
    displayName="Review no approve" creationType="NONE" defaultApproval="true"
pendingBVT="true">
    <description>Update a record and require the user to go through an
    approval process before completing the task.
    </description>
  </taskType>

  <taskType taskTypeId="BeUpdateWithApprovalTask" name="AVOSBeUpdate"
    displayName="Update" creationType="CREATE" pendingBVT="true" displayType="NORMAL">
    <description>Update a record and do not require the user to go through an approval
    process before completing the task. The approval step is optional.
    </description>
  </taskType>
</tasks>
```

Configure Task Assignment for Siperian Workflow Adapter

To configure task assignment for the ActiveVOS workflow adapter based on business entities, use the IDD Configuration Manager to configure task assignment for each subject area. The user can either assign the task directly or allow the Task Manager to assign tasks to users.

1. Log in to the Informatica Data Director Configuration Manager.

   `http://[host]:[port]/bdd/config/`

2. Select the application to update.
3. Click **Edit**.
4. In the Subject Areas tab, select a subject area, and then click **Edit Subject Area**.
5. Click the **Task Assignment** tab, and then click **Add**.
6. From the Task Assignment dialog box, select the task to configure from the Task list.
7. Select the roles and users to which the task can be assigned. Click **OK**.
8. Click **Save**.

9. Click **Generate Business Entity Schema**. Configuration Manager generates the business entity and business entity service configuration.
10. In the MDM Hub, use the Repository Manager to validate the Operational References Store. The Repository Manager validation refreshes the repository data that is cached in the application server.

Configure Tasks

When you switch to the BE ActiveVOS workflow adapter, use the Provisioning Tool to configure tasks to use the BE ActiveVOS workflows.

You can configure the following task properties in the Provisioning tool:

Define who can launch workflows

You configure task triggers so the appropriate ActiveVOS® task workflow is launched after certain events.

Define who can accept tasks

The task type configuration defines which user roles can claim or be assigned ActiveVOS tasks.

Define the default task properties

You can configure task templates so that a task is created with specific properties. For example, you can specify that when a trigger launches a workflow, the tasks have a particular title, priority, due date, and task status.

Setting the Primary and Secondary Workflow Adapters

To migrate to the BE ActiveVOS workflow adapter, select the BE ActiveVOS workflow engine as the primary workflow engine. The name of the workflow adapter for business entities is `BE ActiveVOS`. You can process existing tasks with the secondary workflow engine but you cannot create tasks. Do not select the same workflow engine for the primary and secondary workflow engines.

If you have not previously used Informatica MDM Multidomain Edition with embedded ActiveVOS, select the `BE ActiveVOS` adapter as the primary workflow adapter. You do not need to select a secondary workflow adapter to process existing tasks.

Note: If your IDD configuration is based on subject areas, continue to use the `Informatica ActiveVOS` adapter as the primary workflow engine.

When you add a workflow engine, it becomes the primary workflow engine and the existing primary workflow engine becomes the secondary workflow engine. If you have an existing secondary workflow engine, the workflow engine is dropped from the Operational Reference Store and the tasks are removed from the task inbox.

1. In the Configuration workbench in the Hub Console, click **Workflow Manager**.
2. Acquire a write lock.
3. Select the Workflow Engine tab and ensure that the following BE ActiveVOS workflow adapter information is correct:
   - ActiveVOS Server host
   - ActiveVOS Server port
   - username of trusted user
   - password of trusted user
   - protocol for communication between the MDM Hub and the ActiveVOS Server
4. Select the **Operational Reference Store Workflow Mapping** tab.

   The table in the tab contains all the Operational Reference Store databases in the Hub Store.

5. In the Primary Workflow Engine column, select the workflow engine for the BE ActiveVOS workflow adapter.
6. In the Secondary Workflow Engine column, select a workflow engine.

Processing Existing ActiveVOS Tasks

Processing Existing ActiveVOS Tasks Overview

To work with ActiveVOS tasks that were created before MDM Multidomain Edition version 10.1, routinely run a migration script to populate the tasks with the required presentation parameters. If you do not run the migration script, the tasks do not appear in the Task Manager. Run the migration script until you process all the tasks that were created before you upgraded to version 10.1.

Running the Migration Script

To work with ActiveVOS tasks that were created before MDM Multidomain Edition version 10.1, run the migration script to populate the tasks with the required presentation parameters. If you do not run the migration script, you cannot see the tasks in the Task Manager. Routinely run the script until all the tasks are complete.

Note: To run the script, you can use a properties file. If you do not want to store passwords in a properties file, you can run the script with the properties in the command.

1. Create an MDM Hub super user that belongs to all task management roles.

   The ActiveVOS migration utility requires that you create a super user that belongs to all task management roles.

   Note: After the migration, the tasks are assigned to the same users that the tasks were assigned to before the upgrade.
2. To run the script using a properties file, perform the following steps.
   a. Open the following file in a text editor:

      `<MDM Hub installation directory>\hub\server\bin\build.properties`

   b. Add the following properties to the `build.properties` file.

Property	Description
avos.jdbc.database.driver.jar	Path to the JAR file that contains the JDBC driver for ActiveVOS database. This parameter is populated during the Hub Server installation without the avos prefix in <infamdm installation directory>\conf\avos.install.properties.
avos.jdbc.database.driver.class	JDBC driver class for ActiveVOS database. This parameter is populated during the Hub Server installation without the avos prefix in <infamdm installation directory>\conf\avos.install.properties.
avos.jdbc.database.url	ActiveVOS database connection URL. This parameter is populated during the Hub Server installation without the avos prefix in <infamdm installation directory>\conf\avos.install.properties.

Property	Description
avos.jdbc.database.username	ActiveVOS database user name. This parameter is populated during the Hub Server installation without the avos prefix in <infamdm installation directory>\conf\avos.install.properties.
avos.jdbc.database.password	ActiveVOS database password.
avos.ws.protocol	The protocol for the ActiveVOS server connection. Can be http or https.
avos.ws.host	Host name of the application server where ActiveVOS runs.
avos.ws.port	Port number of the application server connection.
avos.ws.trusted.username	User name of the trusted user. **Note:** The trusted user is created as part of the MDM Multidomain Edition installation and upgrade process.
avos.ws.trusted.password	Password for the trusted user. **Note:** The trusted user is created as part of the MDM Multidomain Edition installation and upgrade process.
avos.hub.username	MDM Hub super user that belongs to all task management roles.
avos.ws.pagesize	Number of tasks processed in one database transaction and batch-loaded from ActiveVOS.
avos.ws.statuses	Optional. Comma-separated list of ActiveVOS task statuses to be processed. For example, READY or IN_PROGRESS. By default all tasks are processed.

c. Open a command prompt.

d. Navigate to the following directory:

- On UNIX. `<MDM Hub installation directory>/hub/server/bin`
- On Windows. `<MDM Hub installation directory>\hub\server\bin`

e. Run the MDM Hub Master Database upgrade script with the following command:

- On UNIX. `sip_ant.sh migrate-avos-sa-tasks`
- On Windows. `sip_ant.bat migrate-avos-sa-tasks`

3. If you want to run the script with the properties in the command, perform the following steps.

a. Open a command prompt.

b. Navigate to the following directory:

- On UNIX. `<MDM Hub installation directory>/hub/server/bin`
- On Windows. `<MDM Hub installation directory>\hub\server\bin`

c. Run the MDM Hub Master Database upgrade script with the properties in the command. For example, you can run the following command:

- On UNIX. `sip_ant.sh migrate-avos-sa-tasks -Davos.jdbc.database.password=!!cmx!! -Davos.ws.protocol=http -Davos.ws.host=localhost -Davos.ws.port=8080 -Davos.ws.pagesize=100 -Davos.ws.trusted.username=avos -Davos.ws.trusted.password=avos -Davos.hub.username=admin`
- On Windows. `sip_ant.bat migrate-avos-sa-tasks -Davos.jdbc.database.password=!!cmx!! -Davos.ws.protocol=http -Davos.ws.host=localhost -Davos.ws.port=8080 -Davos.ws.pagesize=100 -Davos.ws.trusted.username=avos -Davos.ws.trusted.password=avos -Davos.hub.username=admin`

4. Run the script on a regular schedule.
5. After all the tasks for the subject area workflow adapter are processed, you do not need to run the script and you can delete the super user.

Troubleshooting

If you face problems when you migrate from the SA data model in IDD to the BE data model in IDD, use the following information to troubleshoot the issue.

The IDD configuration is invalid

When you generate a business entity schema, sometimes Hierarchy Manager entity base objects do not convert the relationships of Hierarchy Manager-enabled business entities correctly.

To resolve the issue, in the IDD Configuration Manager, edit a subject area with a Hierarchy Manager entity base object. A numeric subtype value indicates an invalid IDD configuration. In the **HM Entity Type** field, select another entity type and then reselect the original entity type until the value in the **Subtype Value** field changes to a nonnumeric value.

Smart search doesn't work

Verify that you completed the following steps:

1. In the Hub Console, you configured smart search and ZooKeeper. You configured only one Process Server as a ZooKeeper server.
2. You set `cmx.ss.enabled=true` in both the `cmxserver.properties` file and the `cmxcleanse.properties` file.
3. If you use IBM WebSphere, you set `pingSolrOnStartup=true` in the `cmxcleanse.properties` file.
4. In the E360 application, you marked fields as searchable in the business entity models.
5. You published the schema with the searchable fields.
6. From the MDM Hub Console, you ran the search index batch job on all the business entities.

A review task opens a business entity in a legacy view, but I turned off the legacy views

The task was migrated from the legacy IDD application. When you open the entity from the task, it opens in the view in which it was edited. When all the legacy tasks are closed, the legacy views are no longer used.

Frequently Asked Questions

Here is a collection of the most frequently asked questions about migrating an IDD application to the BE data model.

The list is sorted into the following categories:

- "Migration FAQs" on page 21
- "IDD Modes FAQs" on page 22

- " ActiveVOS FAQs" on page 23

Migration FAQs

Is it mandatory to upgrade to the business entity data model?

No. Informatica encourages customers to use business entities in IDD, but Informatica still supports customers who want to use the subject area data model. However, you must generate a business entity schema for tasks to work.

Furthermore, after you migrate to business entities, Informatica recommends that you avoid using subject areas entirely. It could cause confusion among data stewards and possibly lead to lost changes to data.

Can I still use user exits with business entities?

No. User exits do not migrate to the Entity 360 framework. However, you can use the external calls in IDD to replace some of the functionality of user exits.

Can I still use the task inbox with business entities?

Yes, you can use the task inbox. The task inbox is located in the Task Manager page, and can also be added to the Start page. The task inbox requires that ActiveVOS is installed and configured.

Is there a downside to migrating to business entity data model in IDD?

There is some loss of functionality between the subject area data model and the business entity data model. For example, it is more difficult to import and export raw data into your system. However, the business entity data model also offers multiple benefits, such as integration with other systems and configurable pages and views. For more information, see "Benefits of the Business Data Model" on page 3 and "Overview" on page 4.

Do Repository Manager changelists contain information about business entities?

Yes. When you export a changelist from the Repository Manager in the Hub Console, the changelist contains all the business entity details. You can also import changelists through the Repository Manager.

Do I need to use ActiveVOS in the business entity data model?

Yes. ActiveVOS is the only supported BPM engine for the business entity data model. For more information about how to configure ActiveVOS, see the *Informatica MDM Multidomain Edition Informatica Data Director - Informatica ActiveVOS Integration Guide*.

How do I use the Provisioning tool?

The following table describes some of the common configurations in the IDD Configuration Manager and their equivalencies in the Provisioning tool.

How do I configure..?	IDD Configuration Manager	Provisioning tool
subject area/ business entity	1. Select an IDD application. 2. On the Subject Areas tab, click **Add Subject Area**.	1. Click **Business Entity** > **Modeling**. 2. Select **Business Entities**.
lookup/reference entity	1. In the Hub Console, use the Schema Manager to configure a base object as a lookup base object. 2. Create a foreign key relationship between a base object and the lookup base object. 3. IDD uses metadata about the foreign key relationship to populate the lookup values.	1. Click **Business Entity** > **Modeling**. 2. Select **Reference Entities**.
relationship	1. In the Hub Console, use the Schema Manager to configure a relationship between base objects. 2. IDD creates a relationship within subject areas based on the relationship between associated base objects.	1. Click **Business Entity** > **Modeling**. 2. Select **Relationships**.
task	1. Select an IDD application. 2. On the Tasks tab, click **Add**.	1. Click **Business Entity > Tasks**. 2. You can define task templates, task types, and triggers.

IDD Modes FAQs

After I upgrade, can I continue to show only the legacy views?

Yes. You do not have to migrate your IDD application to the BE data model.

After you upgrade, if your IDD application does not include BPM workflows, you do not need to perform any of the steps in this guide.

If your IDD application includes BPM workflows, you must generate the business entity schema. Informatica Data Director requires that the schema exist. You can continue to use the same workflow engine. You do not need to change the IDD mode, nor do you need to use the Provisioning tool.

Note: In this mode, the features and views that are based on the BE data model are not available.

To continue to use SA data model, legacy views, and workflow engine, generate the business entity schema:

1. "Prepare the ORS and the IDD Application" on page 9
2. "Generate the Business Entity Schema" on page 10

Can I show the legacy views and the E360 views at the same time?

Yes, you can choose to show both sets of views. This is called the Hybrid mode. For example, you might want to use the Hybrid mode temporarily while you design the Start page and Entity view layouts.

Caution: In the Hybrid mode, if you need to make a change to the schema, you must make the change twice: once in the IDD Configuration Manager and again in the Provisioning tool. If you regenerate a schema from the IDD Configuration Manager and publish the changes in the Provisioning tool, any customizations that you made to the migrated elements are overwritten. New elements, such as a business entity model with a different name than a subject area, are preserved because they are not overwritten.

To use the Hybrid mode, perform all the steps in this document. When you change the mode, use the following settings:

Properties	Hybrid Mode
cmx.dataview.enabled	true
cmx.e360.view.enabled	true
cmx.e360.match_xref.view.enabled	true
cmx.ss.enabled	true

In the Provisioning tool, you must create an IDD E360 application with the same name as the legacy IDD application.

ActiveVOS FAQs

If I use the Subject Area-based ActiveVOS workflow adapter, which view do I use to create merge and unmerge tasks?

If you use the Subject Area-based ActiveVOS workflow adapter, use the legacy Matches view and the legacy XREF view. Set `cmx.e360.match_xref.view.enabled` to `false` to use these legacy views to create tasks. When you upgrade to version 10.2, `cmx.e360.match_xref.view.enabled` is `false` by default.

How do you process existing merge tasks?

Any merge tasks created prior to version 10.2 open in the legacy Match view. If you enable the Entity 360 framework, merge tasks that are created in version 10.2 are opened in the Entity 360 framework Matching Records view.

How do you process unmerge tasks?

All unmerge tasks open in the Entity 360 framework Cross-reference Records view, regardless of the `cmx.e360.match_xref.view.enabled` setting.

How do you process review tasks?

All review tasks are processed in the Entity 360 framework Entity view.

If I use the business entity ActiveVOS workflow adapter, which task inbox should I use?

Use the Entity 360 framework Task Manager and task inbox in the Informatica Data Director user interface configuration for the Start page and the Entity view. If you use the legacy task inbox with the business entity ActiveVOS workflow adapter, an error occurs.

Glossary of Terms

business entity

An entity with significance to an organization. Organizations commonly define business entity types to represent customers, suppliers, employees, products, and accounts. For example, a business entity type might be Person. The customer John Smith is a business entity of type Person.

In the business entity data model of Informatica Data Director, business entities replace subject areas.

business entity services

A set of operations that run MDM Hub code to create, update, delete, and search for base object records in a business entity. You can develop a custom user interface that can run Java code or JavaScript code to make business entity service calls.

business entity view

A view that represents a condensed version of a business entity.

Entity 360 framework

Business entity-based Informatica Data Director (IDD). With the Entity 360 framework enabled, IDD users edit and manage master data from an entity workspace instead of the Data workspace.

IDD Configuration Manager

A web-based utility to add, change, and manage Informatica Data Director applications that are based on the subject area data model.

lookup

In Informatica Data Director, a dropdown list of values from which the application user can choose. Usually, lookup values are defined in a physical lookup base object table with a foreign key between the base object and the lookup base object.

Provisioning tool

A web-based utility to define business entity models, tasks, and transformations, and design the user interface for Informatica Data Director.

In the business entity data model of Informatica Data Director, the Provisioning tool replaces the IDD Configuration Manager.

reference entity

A business entity that is associated with a lookup base object.

In the business entity data model of Informatica Data Director, reference entities replace lookups.

subject area

In Informatica Data Director, a collection of data that can be treated as a unit from a business perspective.

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